GA-04 | Basics of Environment & Pollution Control

RRB JE CBT-2 (CEN 05/2025) — General Abilities | Weightage: 10 Questions

> **Exam angle:** RRB asks straightforward, high-yield questions on ecosystem concepts, pollution types & control devices, solid-waste classification, global issues (global warming, ozone, acid rain), Indian environmental movements, and the legislation timeline. Every sub-topic seen in actual 2025 CBT-2 papers is covered, with a PYQ Corner, 25 practice MCQs, and rapid-revision tables.

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1. Ecosystem Basics

An **ecosystem** is a self-sustaining unit where living organisms interact with each other and their physical surroundings. Term coined by **A.G. Tansley (1935)**.

1.1 Components

Component	Meaning	Examples
Biotic	Living components	Plants (producers), animals (consumers), microbes (decomposers)
Abiotic	Non-living components	Sunlight, temperature, water, air, soil, minerals

Biotic sub-groups:

Group	Role	Example
Producers (autotrophs)	Make food via photosynthesis	Green plants, algae
Consumers (heterotrophs)	Depend on others for food	Herbivores, carnivores, omnivores
Decomposers (saprotrophs)	Break down dead matter, recycle nutrients	Bacteria, fungi

1.2 Food Chain, Food Web & Trophic Levels

- **Food chain:** linear sequence of "who eats whom" — energy flows from one level to the next.

Example: *Grass → Grasshopper → Frog → Snake → Hawk*.

- **Food web:** many interconnected food chains — more realistic and stable.
- **Trophic levels:** feeding positions in a food chain.

Trophic Level	Organism	Example
T1	Producer	Grass
T2	Primary consumer (herbivore)	Grasshopper
T3	Secondary consumer (carnivore)	Frog
T4	Tertiary consumer	Snake
T5	Top / apex consumer	Hawk

1.3 Energy Flow — The 10% Law ★

- Energy flow in an ecosystem is **unidirectional** (one-way, from Sun → producers → consumers).
- **Lindeman's 10% Law:** only about **10% of energy** is transferred from one trophic level to the next; the remaining ~90% is lost as heat (metabolism, respiration). This is why food chains are usually limited to 4–5 links.
- **Ecological pyramids:** number, biomass, or energy. The **pyramid of energy is always upright**.

1.4 Biomagnification & Bioaccumulation ★

Term	Meaning
Bioaccumulation	Build-up of a toxic substance **within a single organism** over its lifetime (faster intake than excretion)
Biomagnification	Progressive **increase in concentration of a toxic, non-degradable substance at each higher trophic level** of a food chain

- Classic example: **DDT** (a pesticide) in **aquatic ecosystems** — water has a low DDT concentration, but it magnifies through plankton → small fish → large fish → **fish-eating birds**, where it is highest and causes thin egg-shells. Other biomagnifying pollutants: **mercury (Minamata disease), lead, arsenic**.
- These substances biomagnify because they are **fat-soluble and non-biodegradable**, so they are stored, not excreted.

> **2025 CBT-2 asked:** In which ecosystem is biomagnification most significantly observed? → **Aquatic ecosystems**, because pollutants like DDT and mercury enter water and concentrate up long aquatic food chains.

1.5 Eutrophication

- **Eutrophication** = over-enrichment of a water body with **nutrients (nitrates & phosphates)** from fertilizer runoff and sewage.
- Leads to **algal bloom** → blocks sunlight → algae die → decomposers consume oxygen → **dissolved oxygen falls** → fish and aquatic life die. An **algal bloom of toxic cyanobacteria** can poison the water.

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2. Biodiversity & Conservation

- **Biodiversity** = variety of life — genetic, species, and ecosystem diversity. Term popularised by **E.O. Wilson**.
- **Biodiversity hotspots in India (4):** (i) **Himalayas**, (ii) **Western Ghats**, (iii) **Indo-Burma region**, (iv) **Sundaland (Nicobar Islands)**.

Conservation methods:

Type	Meaning	Examples
In-situ	Protecting species **in their natural habitat**	National parks, wildlife sanctuaries, biosphere reserves, tiger reserves
Ex-situ	Protecting species **outside their natural habitat**	Zoos, botanical gardens, gene/seed banks, aquariums

- India has 100+ national parks and 18 biosphere reserves. **Project Tiger** (1973) and **Project Elephant** (1992) are flagship in-situ programmes.
- **IUCN Red List** categories: Extinct → Critically Endangered → Endangered → Vulnerable → Near Threatened → Least Concern.

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3. Air Pollution

Air pollution = presence of harmful substances in the atmosphere beyond safe limits.

3.1 Major Pollutants, Sources & Effects

Pollutant	Main Source	Effect
Carbon monoxide (CO)	Incomplete fuel combustion, vehicles	Reduces oxygen carriage in blood; can be fatal
Carbon dioxide (CO_2)	Fossil-fuel burning	Major greenhouse gas → global warming
Sulphur dioxide (SO_2)	Coal burning, thermal plants	Respiratory problems; **acid rain**
Oxides of nitrogen (NO_x)	Vehicles, industry	Acid rain, smog
Particulate matter (PM2.5, PM10)	Dust, vehicles, construction	Lung diseases, reduced visibility
CFCs	Refrigerants, aerosols	Ozone-layer depletion
Lead	Old petrol, batteries	Nervous-system damage
Ground-level ozone (O_3)	Photochemical reactions	Photochemical smog, breathing issues

Pollution classification: *Primary pollutants* are emitted directly (CO, SO_2, NO_x); *secondary pollutants* form by reaction in the atmosphere (ozone, smog, PAN).

3.2 Smog Types

Type	Where/When	Cause
Classical / London smog	Cool, humid; winter	Smoke + fog + SO_2 (reducing smog)
Photochemical / Los Angeles smog	Warm, sunny; day	NO_x + hydrocarbons + sunlight → ozone, PAN (oxidising smog)

3.3 Control Devices ★

Device	Function
Electrostatic Precipitator (ESP)	Removes **particulate matter** from industrial flue gases using electric charge; efficiency up to 99%
Scrubber	Removes **gases (SO_2) and dust** by passing flue gas through a liquid spray
Catalytic converter	Fitted in vehicle exhaust; converts CO → CO_2, NO_x → N_2, unburnt hydrocarbons → CO_2+H_2O (uses platinum/palladium; needs unleaded petrol)
Bag filter / Cyclone separator	Trap dust particles

- **AQI (Air Quality Index):** 0–50 Good, 51–100 Satisfactory, 101–200 Moderate, 201–300 Poor, 301–400 Very Poor, 401–500 Severe.

4. Water Pollution

Sources: domestic sewage, industrial effluents, agricultural runoff (fertilizers/pesticides), oil spills, thermal pollution, plastic waste.

Parameter	Full form	Meaning
BOD	Biochemical Oxygen Demand	Oxygen needed by microbes to decompose organic matter; **high BOD = more pollution**
COD	Chemical Oxygen Demand	Oxygen needed to chemically oxidise all matter; always ≥ BOD
DO	Dissolved Oxygen	Oxygen available to aquatic life; **low DO = polluted**
pH	—	Acidity/alkalinity; safe drinking water ≈ 6.5–8.5

Sewage treatment stages:

Stage	Process
Primary	Physical removal — screening, sedimentation of solids
Secondary	Biological — microbes break down organic matter (aeration tanks)
Tertiary	Chemical — removes remaining nutrients/pathogens; disinfection (chlorination)

Ganga cleaning: Namami Gange programme (2014, integrated conservation mission; replaced the earlier Ganga Action Plan of 1985) — sewage treatment, riverfront development, afforestation, biodiversity. Nodal: National Mission for Clean Ganga (NMCG).

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5. Soil & Noise Pollution

Soil pollution — causes: excess chemical fertilizers & pesticides, industrial waste, plastics, mining, acid rain. **Effects:** loss of fertility, groundwater contamination, reduced crop yield. **Control:** organic farming, crop rotation, afforestation, reduced plastic use.

Noise pollution: unwanted/harmful sound, measured in **decibels (dB)**.

Situation	Approx. level
Safe/normal conversation	~ 60 dB
WHO safe daytime limit (residential)	~ 55 dB (day), 45 dB (night)
Industrial safe limit (India)	75 dB (day) / 70 dB (night)
Pain threshold	~ 120–130 dB

Effects: hearing loss, hypertension, sleep disturbance, stress. Control: green belts, silencers, sound barriers, restricting horns/loudspeakers.

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6. Solid Waste Management ★★

6.1 Sources of Municipal Solid Waste (MSW)

Municipal solid waste is the everyday, non-hazardous rubbish generated from:

Source	Typical waste
Homes / residences	Kitchen scraps, packaging, paper, plastic
Schools & offices	Paper, stationery, general litter
Shops, markets, restaurants	Packaging, food waste
Hospitals (general/non-clinical areas)	Office paper, canteen waste, packaging
Streets & public places	Sweepings, garden waste

> **Important exam point: Radioactive waste and industrial hazardous waste are NOT part of municipal solid waste.** They are collected, treated and disposed of **separately** under special rules. Biomedical (clinical) waste from hospitals is also handled separately, not as ordinary MSW.

> **2025 CBT-2 asked:** Which of the following is NOT a source of municipal solid waste? → **Radioactive waste** (it comes from nuclear plants/labs and is managed under a separate radioactive-waste-disposal system, not with municipal garbage).

6.2 Classification of Waste

Category	Meaning	Examples
Biodegradable	Decomposed by microbes	Food waste, paper, leaves, cotton
Non-biodegradable	Not broken down naturally	Plastic, glass, metal, polythene
Hazardous	Toxic/flammable/corrosive/reactive	Chemicals, batteries, paints
Biomedical	From hospitals/clinics	Syringes, dressings, body fluids (colour-coded bins)
E-waste	Discarded electronics	Old phones, computers, TVs
Radioactive	From nuclear activity	Handled separately (never in MSW)

Colour-coded biomedical waste bins (India):

Colour	Waste type
Yellow	Anatomical/soiled waste, incinerated
Red	Contaminated recyclable plastic (tubing, catheters)
White (translucent)	Sharps (needles, blades)
Blue	Glassware, metallic implants

6.3 Waste Management Methods — the 5 R's & Disposal

- **5 R's: Refuse, Reduce, Reuse, Recycle, Recover** (also 3R: Reduce, Reuse, Recycle).

Method	Description	Note
Composting	Biodegradable waste decomposed into manure by microbes	Best for organic waste; vermicomposting uses earthworms
Recycling	Reprocessing waste into new products	Paper, glass, metal, plastic
Landfill	Waste buried in engineered pits with liners	Sanitary landfill prevents groundwater contamination; produces methane
Incineration	Controlled burning at high temperature	Reduces volume; used for biomedical/hazardous waste; can generate energy but emits gases
Waste-to-energy	Combustion/biogas produces electricity	—

7. Radioactive Pollution

- **Sources:** nuclear power plants, nuclear weapon tests, uranium mining, medical X-rays/radiotherapy, radioactive isotopes in labs, nuclear accidents (Chernobyl 1986, Fukushima 2011).
- **Effects:** ionising radiation damages DNA → **mutations, cancer, genetic disorders, radiation sickness, birth defects**. High doses are fatal.

> **2025 CBT-2 asked:** Radiation from nuclear waste primarily causes what biological effect? → **Mutations** (changes in DNA/genes), which can lead to cancer and hereditary defects.

- **Disposal:** radioactive waste is **isolated and stored in shielded containers deep underground (geological disposal)** or in sealed concrete/steel casks; it is **never** mixed with municipal waste. Classified as low-, intermediate-, and high-level waste by activity.

8. Global Environmental Issues

8.1 Greenhouse Effect & Global Warming ★

- **Greenhouse effect:** greenhouse gases in the atmosphere trap outgoing infrared (heat) radiation, warming the Earth — a natural process that keeps the planet habitable. Excess gases **enhance** it, causing **global warming**.

Greenhouse Gas	Main Source	Note
Carbon dioxide (CO_2)	Fossil-fuel burning, deforestation	Largest overall contributor
Methane (CH_4)	Paddy fields, cattle, landfills	~25× more potent than CO_2
Nitrous oxide (N_2O)	Fertilizers	Very potent
CFCs / HFCs	Refrigerants	Also deplete ozone
Water vapour	Natural	Most abundant

- **Global warming:** global average surface temperature has risen by **about 1 °C (≈1.1 °C)** above pre-industrial levels *(as of the mid-2020s, IPCC)*. The Paris Agreement targets keeping the rise **well below 2 °C, ideally 1.5 °C**.
- **Consequences:** melting glaciers & polar ice, sea-level rise, extreme weather, crop failure, species loss.

> **2025 CBT-2 asked:** Global warming refers to what? → the **rise in Earth's average temperature** due to increased greenhouse gases; the observed rise is about **1 °C** over pre-industrial times *(as-of mid-2020s)*.

8.2 Ozone-Layer Depletion

- **Ozone layer** (in the **stratosphere**) absorbs harmful **ultraviolet (UV)** radiation from the Sun.
- Depleted mainly by **CFCs (chlorofluorocarbons)** and halons → an "**ozone hole**" over Antarctica.
- Effects: skin cancer, cataracts, weakened immunity, crop damage.
- **Montreal Protocol (1987)** — international treaty to phase out ozone-depleting substances (very successful). **World Ozone Day: 16 September**.

8.3 Acid Rain

- **Acid rain** = rain with **pH below 5.6**, caused by **SO_2 and NO_x** reacting with water vapour to form sulphuric and nitric acids.
- Effects: damages buildings/monuments (**e.g. Taj Mahal — "marble cancer"**), acidifies soil and lakes, harms forests and aquatic life.

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9. Environmental Movements in India ★

Movement	Year	Place	Leader(s)	About
Chipko Movement	1973	**Uttarakhand** (then UP)	**Sunderlal Bahuguna**, Chandi Prasad Bhatt, Gaura Devi	Villagers **hugged trees** to prevent felling; "chipko" = to embrace; slogan "ecology is permanent economy"
Appiko Movement	1983	Karnataka (Western Ghats)	Pandurang Hegde	Southern counterpart of Chipko ("appiko" = to hug in Kannada)
Silent Valley Movement	1970s–80s	Kerala	M.K. Prasad, activists	Saved the Silent Valley tropical rainforest from a hydel dam; declared a National Park (1984)
Narmada Bachao Andolan	1985	Gujarat/MP/Maharashtra	**Medha Patkar**, Baba Amte	Against big dams on the Narmada (Sardar Sarovar); rehabilitation of displaced people
Bishnoi Movement	1730	Rajasthan	Amrita Devi Bishnoi	Earliest — sacrificed lives to protect Khejri trees

> **2025 CBT-2 asked:** The Chipko Movement is associated with which state / what activity? → **Uttarakhand**; villagers hugged trees to stop deforestation. Leader: **Sunderlal Bahuguna**.

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10. Indian Environmental Legislation — Timeline Table ★

Year	Act / Body	Purpose
1972	**Wildlife (Protection) Act**	Protects wild animals, birds, plants; sets up national parks & sanctuaries
1974	**Water (Prevention & Control of Pollution) Act**	Prevents water pollution; created **CPCB & SPCBs**
1980	Forest (Conservation) Act	Restricts diversion of forest land
1981	**Air (Prevention & Control of Pollution) Act**	Controls air pollution
1986	**Environment (Protection) Act ★**	Umbrella / most comprehensive law; passed after the **Bhopal Gas Tragedy (1984)**; empowers Central Govt to protect and improve the environment
1988	Motor Vehicles Act	Vehicle emission norms
2002	Biological Diversity Act	Conservation & sustainable use of biodiversity
2010	**National Green Tribunal (NGT) Act**	Special court for speedy disposal of environmental cases

> **2025 CBT-2 asked:** Which Act is the umbrella legislation for environmental protection in India, enacted after the Bhopal disaster? → **The Environment (Protection) Act, 1986.**

- **Pollution Control Boards: CPCB** (Central Pollution Control Board, national) and **SPCBs** (State Pollution Control Boards) set standards, monitor and enforce pollution norms. CPCB works under the Ministry of Environment, Forest & Climate Change (MoEFCC).

11. International Conventions — One-Liners

Event	Year	About
Stockholm Conference	1972	First UN environment conference; **World Environment Day, 5 June** originates here; led to UNEP
Rio Earth Summit (UNCED)	1992	Agenda 21; Convention on Biological Diversity; UNFCCC framework
Kyoto Protocol	1997	First binding targets to cut greenhouse-gas emissions
Ramsar Convention	1971	Conservation of **wetlands**
CITES	1973	Regulates trade in endangered species
Montreal Protocol	1987	Phase-out of ozone-depleting substances
Paris Agreement	2015	Limit global warming well below 2 °C (aim 1.5 °C); nationally determined contributions
COP (Conference of Parties)	Annual	UNFCCC climate meetings; recent: COP28 (Dubai, 2023), COP29 (Baku, 2024) *(as of 2024–25)*

12. PYQ Corner — Actual RRB JE CBT-2 2025 Questions Solved

Q1. Which of the following is NOT a source of municipal solid waste?

Answer: Radioactive waste. MSW comes from homes, schools, offices, shops and general hospital areas. Radioactive waste (from nuclear plants/labs) and industrial hazardous/biomedical waste are managed separately under special disposal rules.

Q2. Which Act is the comprehensive umbrella legislation for environmental protection in India?

Answer: The Environment (Protection) Act, 1986, enacted after the 1984 Bhopal Gas Tragedy; empowers the Central Government to take all measures to protect and improve the environment.

Q3. Radiation from nuclear/radioactive waste primarily causes which biological effect?

Answer: Mutations — ionising radiation damages DNA, causing genetic mutations that can lead to cancer, hereditary defects and radiation sickness.

Q4. Global warming refers to what, and by roughly how much has temperature risen?

Answer: Global warming is the **rise in Earth's average surface temperature** due to increased greenhouse gases. The rise is about **1 °C (≈1.1 °C)** above pre-industrial levels *(as of the mid-2020s, IPCC)*.

Q5. The Chipko Movement originated in which state and involved what activity?

Answer: Uttarakhand (then part of Uttar Pradesh). Villagers, led by **Sunderlal Bahuguna** and Chandi Prasad Bhatt, **hugged trees** to prevent them from being cut.

Q6. Biomagnification is most significantly observed in which ecosystem?

Answer: Aquatic ecosystems. Fat-soluble, non-biodegradable pollutants such as **DDT and mercury** concentrate up long aquatic food chains, reaching highest levels in top predators like fish-eating birds.

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13. Practice MCQs (25) — Answers at the End

1. The term "ecosystem" was coined by: (a) Odum (b) Tansley (c) Wilson (d) Lindeman
2. Green plants in a food chain are: (a) Consumers (b) Decomposers (c) Producers (d) Predators
3. According to the 10% law, energy transferred to the next trophic level is: (a) 1% (b) 10% (c) 50% (d) 90%
4. Biomagnification of DDT is highest in: (a) Water (b) Plankton (c) Small fish (d) Fish-eating birds
5. Eutrophication is caused by excess: (a) Oxygen (b) Nitrates and phosphates (c) Carbon dioxide (d) Chlorine
6. Which is a biodiversity hotspot in India? (a) Thar Desert (b) Western Ghats (c) Deccan Plateau (d) Gangetic Plain
7. Protecting a species in a zoo is: (a) In-situ (b) Ex-situ (c) On-site (d) Habitat conservation
8. Electrostatic precipitators remove: (a) Gases (b) Particulate matter (c) Bacteria (d) Odour
9. A catalytic converter is fitted in: (a) Chimneys (b) Vehicle exhaust (c) Sewage plants (d) AC units
10. High BOD in water indicates: (a) Clean water (b) More pollution (c) High oxygen (d) Low bacteria
11. Secondary sewage treatment is mainly: (a) Physical (b) Chemical (c) Biological (d) Thermal
12. Namami Gange is related to cleaning the river: (a) Yamuna (b) Ganga (c) Godavari (d) Narmada
13. Noise is measured in: (a) Hertz (b) Decibels (c) Lumens (d) Pascals
14. Which is NOT municipal solid waste? (a) Kitchen waste (b) Paper (c) Radioactive waste (d) Packaging
15. Composting is best suited for: (a) Plastic (b) Glass (c) Biodegradable waste (d) Metal
16. White (translucent) biomedical bins are for: (a) Anatomical waste (b) Sharps (c) Plastic (d) Glass
17. The gas mainly responsible for global warming is: (a) O_2 (b) CO_2 (c) N_2 (d) He

18. Global temperature has risen by about: (a) 0.1 °C (b) 1 °C (c) 5 °C (d) 10 °C
19. Ozone layer is depleted mainly by: (a) CO_2 (b) CFCs (c) CH_4 (d) SO_2
20. The Montreal Protocol (1987) deals with: (a) Global warming (b) Ozone depletion (c) Wetlands (d) Biodiversity
21. Acid rain has a pH: (a) Above 7 (b) Exactly 7 (c) Below 5.6 (d) Above 8
22. The Chipko Movement is associated with: (a) Rajasthan (b) Uttarakhand (c) Kerala (d) Karnataka
23. The Environment Protection Act was passed in: (a) 1974 (b) 1981 (c) 1986 (d) 2010
24. CPCB stands for: (a) Central Public Control Board (b) Central Pollution Control Board (c) City Pollution Control Body (d) Central Power Control Board
25. World Environment Day is observed on: (a) 22 April (b) 5 June (c) 16 September (d) 2 December

Answers: 1-b, 2-c, 3-b, 4-d, 5-b, 6-b, 7-b, 8-b, 9-b, 10-b, 11-c, 12-b, 13-b, 14-c, 15-c, 16-b, 17-b, 18-b, 19-b, 20-b, 21-c, 22-b, 23-c, 24-b, 25-b

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14. Rapid-Revision Tables

14.1 Pollution Control Devices

Device	Removes	Used in
Electrostatic precipitator (ESP)	Particulate matter	Thermal plants, industry
Scrubber	SO_2 gas + dust	Industrial chimneys
Catalytic converter	CO, NO_x, hydrocarbons	Vehicle exhaust
Bag/cyclone filter	Dust particles	Factories

14.2 Global Issues Snapshot

Issue	Cause	Key treaty / fact
Global warming	CO_2, CH_4 (greenhouse gases)	~1 °C rise; Paris 2015, Kyoto 1997
Ozone depletion	CFCs	Montreal Protocol 1987; Ozone Day 16 Sept
Acid rain	SO_2, NO_x	pH < 5.6; damages Taj Mahal

14.3 Environmental Days

Day	Date
World Environment Day	5 June
Earth Day	22 April
World Ozone Day	16 September
World Water Day	22 March
World Wildlife Day	3 March

14.4 Legislation in One Line

Act	Year
Wildlife Protection	1972
Water Act	1974
Air Act	1981
Environment Protection Act	1986
NGT	2010

14.5 Movements in One Line

Movement	State	Cause
Chipko	Uttarakhand	Save trees (hug)
Appiko	Karnataka	Save Western Ghats forests
Silent Valley	Kerala	Save rainforest from dam
Narmada Bachao	MP/Gujarat	Against big dams

End of GA-04. Next: GA-05 General Awareness capsule.